Quality Management

11 — Image processing for inspection: from pixels to a decision

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Today

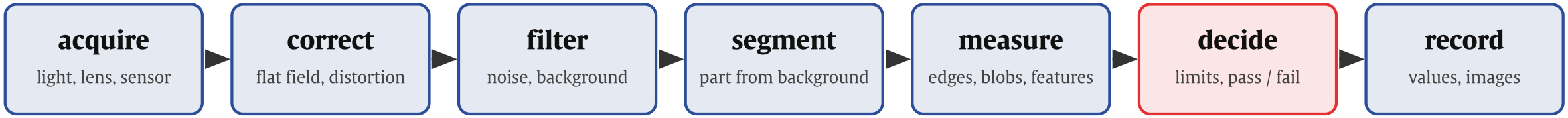
The image is a matrix of numbers. Now: what to do with them

Light, lens and sensor give an image. Image processing turns it into **a measurement and a decision**: accept or reject. Every step does one simple thing to the numbers — and every step can add an error.

1. **The pipeline**
   and the histogram
2. **Filtering**
   noise, background, the frequency view
3. **Edges**
   where the grey level changes
4. **Parts and defects**
   threshold, clean-up, blobs
5. **Finding and deciding**
   matching, codes, pass or fail, validation

The inspection pipeline

The same steps in almost every vision inspection



Two cases we follow

Measuring: the washers of the lab — outer and hole diameter, against ISO 7089.

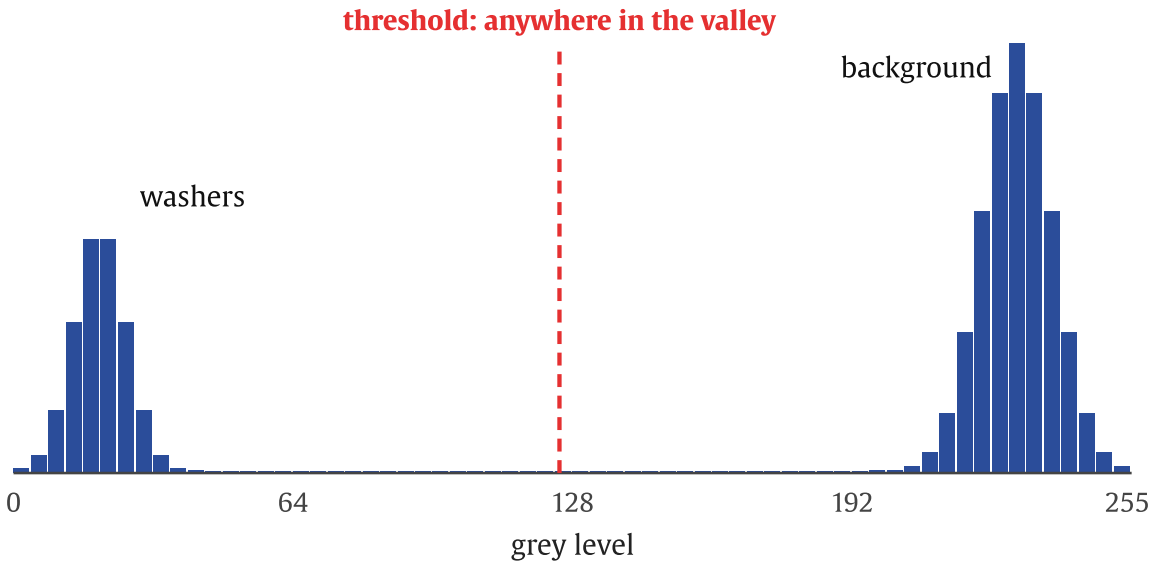
Finding defects: scratches on a machined steel cover, under dark-field light (lecture 10).

Each step is part of the gauge

Correcting, filtering, thresholding all move numbers, and some move edges. They belong in the measurement's uncertainty and in its validation, like the light and the lens.

The histogram: look at the numbers first

How many pixels have each grey level



- The histogram forgets *where* the pixels are and keeps only *how many* have each value. For an 8-bit image: 256 counts.
- A backlit washer gives two peaks: the parts near 20, the background near 230. Far apart and narrow means **high contrast and low noise**.
- **Exposure check:** a spike at 255 means saturation, a spike at 0 means clipping (lecture 9). Neither may be there.
- **Threshold:** any value in the empty valley between the peaks separates part from background.

> **Point operations**
>
> Each grey level changed through a *look-up table*: contrast stretch, inversion, gamma. They change how the image looks, not what it contains.

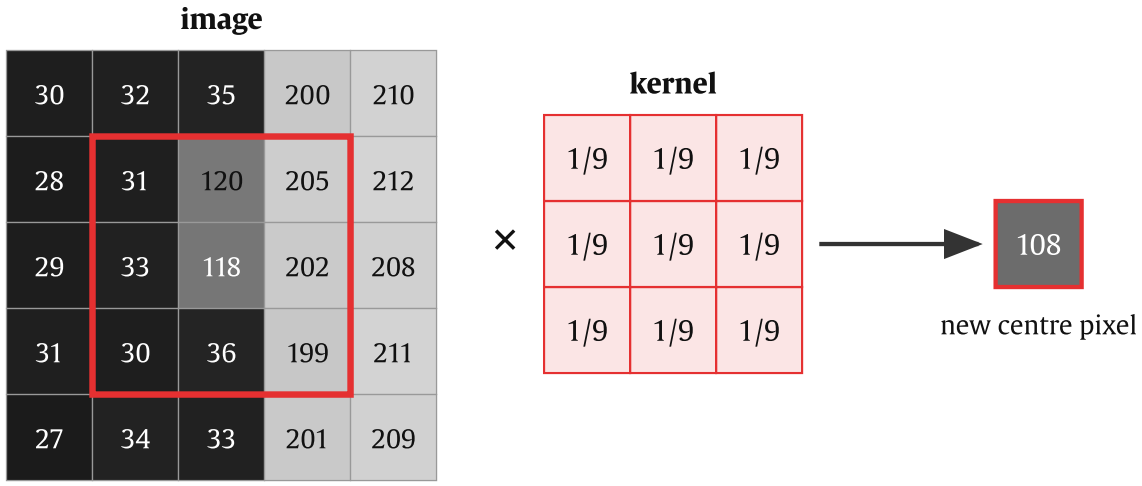
PART 1

Filtering

Neighbourhoods · Smoothing · Background · The frequency view

One operation behind most filters

Replace each pixel by a weighted sum of its neighbours



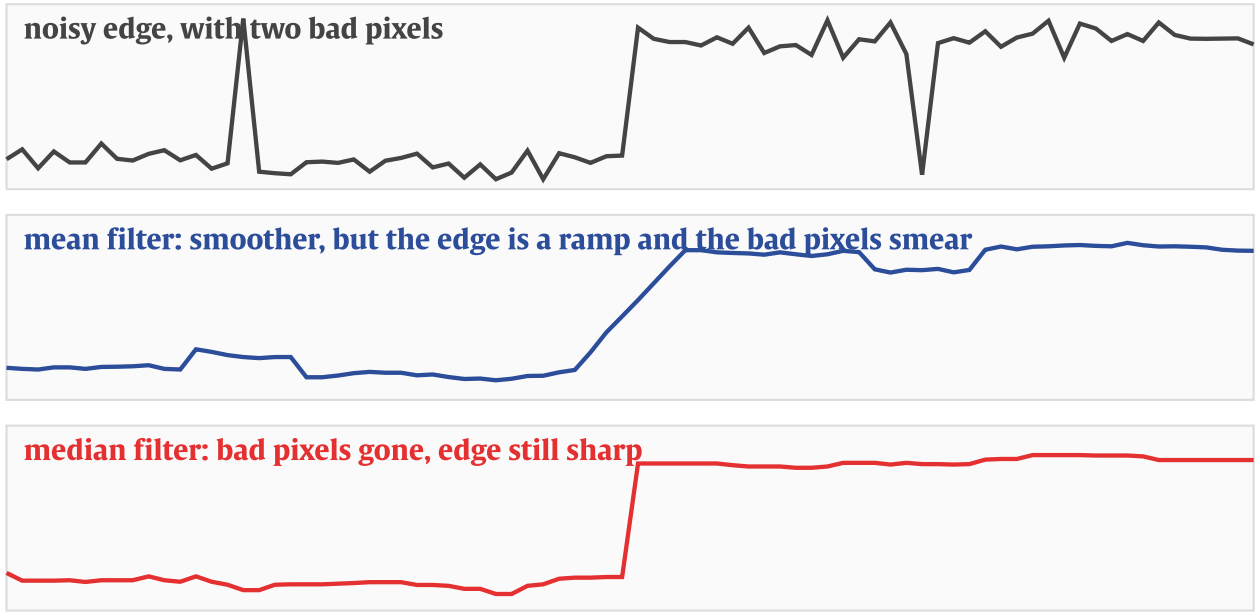
the kernel moves on by one pixel, and the sum is repeated everywhere

- A small grid of weights — the **kernel**, often 3 × 3 — slides over the image. At every position, each pixel under it is multiplied by its weight, and the sum becomes the new pixel.
- This is **convolution**. One operation, and the kernel decides what it does:

Kernel	Effect
all weights equal	averages: **smooths**
large in the centre, falling off	Gaussian: smooths gently
−1, 0, +1 across a direction	differences: **finds edges**
centre minus neighbours, added back	**sharpens**

Smoothing against noise

Less noise, softer edges: choose the smallest filter that does the job



- **Mean (box) filter:** the average of the neighbourhood. Random noise falls as $\sqrt{k}$ for k pixels averaged (lecture 9) — and the edge becomes a ramp.
- **Gaussian filter:** closer neighbours weigh more. Its width σ sets how strong it is. The standard smoothing.
- **Median filter:** the middle value of the neighbourhood, not an average. Single wrong pixels — a hot pixel, a speck of dust — disappear, and edges stay sharp.

For measurement

It does not move a straight edge, but it rounds corners, shrinks small features and merges close ones. Use the same filter when calibrating and measuring.

Uneven background and sharpening

Two more uses of the same idea

Removing an uneven background

Light that is brighter in the centre (lecture 10) makes a fixed threshold fail at the edges. Smooth the image very strongly — only the slow background survives — and **subtract** it: what remains is the detail on a flat background. Better still, fix the light, or correct with a flat field (lecture 9).

Sharpening

Subtracting a smoothed copy leaves only the fine detail; adding it back, amplified, makes edges crisper (*unsharp masking*). It helps people see a defect — and it amplifies the noise and creates overshoots next to edges. **Do not sharpen images you measure.**

The rule behind both

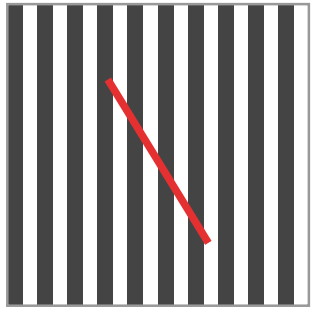
Processing can rearrange what the image contains; it cannot add information the light did not put there. A step that makes the image look better is not automatically a step that makes the measurement better — true also of advanced denoising (wavelets, neural networks): excellent for looking, to be validated before measuring.

The frequency view, without the mathematics

An image as a sum of stripe patterns, coarse to fine

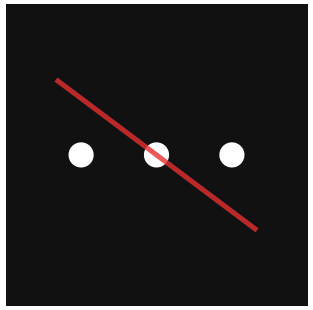
Every image can be written as a sum of stripe patterns of different fineness, direction and strength — its **spectrum** (Fourier, 1807; computed with the fast Fourier transform, FFT). Coarse stripes are **low frequencies**: smooth shading, uneven light. Fine stripes are **high frequencies**: edges, fine texture, and noise.

- Smoothing removes high frequencies: a **low-pass** filter. Sharpening boosts them.
- Filtering in the image is the same as weighting the spectrum — two views of one operation.



texture with a scratch

→



spectrum: stripes = 3 points

1 · Defects on regular textures

Woven fabric, wire mesh, turned grooves, wafer patterns: the regular texture is a few bright points in the spectrum. Remove them, transform back, and only what breaks the pattern is left — the defect.

2 · How sharp a lens is: the MTF

Lens datasheets give the **modulation transfer function**: the contrast the lens keeps for stripes of each fineness, in line pairs per millimetre. It is how lenses are compared and bought.

3 · Aliasing, again

Stripes finer than two pixels come back as false coarse ones (lecture 8). Many cameras put a weak optical low-pass filter in front of the sensor to prevent it.

PART 2

Edges

What makes an edge · Finding it · Where it really is

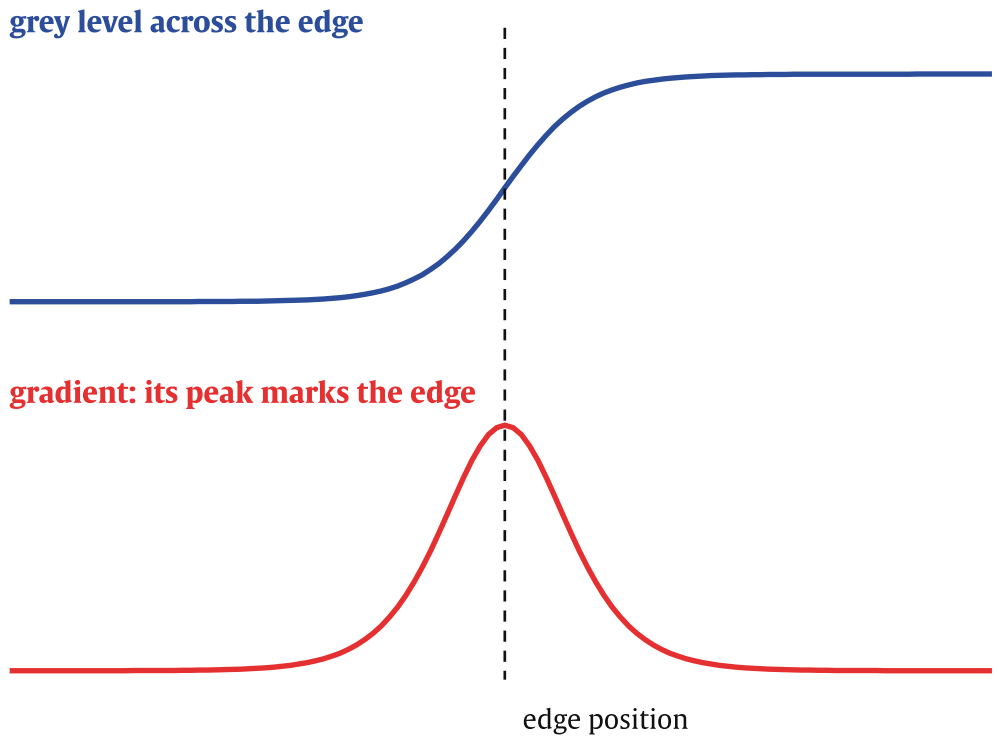
What makes an edge

A rapid change of grey level — not always where the part ends

An edge is where the grey level changes quickly across a few pixels. Four physical causes produce one:

Cause	Example	Wanted?
the part ends (depth jump)	outline of a washer	**yes**
the surface turns	a chamfer, a fold	often
the material changes	print, paint, a stain	sometimes
the light changes	**a shadow, a reflection**	**no**

Software cannot tell a shadow's edge from a part's edge. The light can: that is why it is chosen first.



Finding edges

Differences between neighbours, as a kernel

- The change of grey level from one pixel to the next is the **gradient**: large at an edge, zero on flat areas. Computed with a kernel of differences, in x and in y.
- The **Sobel** kernel differentiates across the edge and averages along it, so it is less sensitive to noise. From the two directions: the edge's **strength** and its **direction**.
- Differences amplify noise, so images are **smoothed first** — usually with a Gaussian. Smoothing and differentiating combine into one kernel: the derivative of a Gaussian.
- Keep the edges whose strength exceeds a threshold. With two thresholds — strong edges, plus weak ones connected to them — broken edges are avoided.

Sobel, across x		
−1	0	+1
−2	0	+2
−1	0	+1

Better than a pixel

The grey profile across an edge spreads over a few pixels. Interpolating it — where it crosses half-way, or where the gradient peaks — places the edge to a fraction of a pixel. How far that can be trusted, and how to calibrate it, is lecture 12.

Recap TAKEAWAYS

The pipeline, filtering and edges

1. **The pipeline:** acquire, correct, filter, segment, measure, decide, record. Every step is part of the gauge.
2. **The histogram first:** exposure, contrast, noise and the threshold, in one plot.
3. **Convolution:** a weighted sum of neighbours. The kernel decides: smooth, find edges, sharpen. The median is the robust alternative for single bad pixels.
4. **The frequency view:** smoothing is low-pass. Useful for periodic textures, for reading a lens's MTF, and for understanding aliasing.
5. **Edges:** gradient after smoothing. A shadow makes an edge too — the light, not the software, prevents it.

PART 3

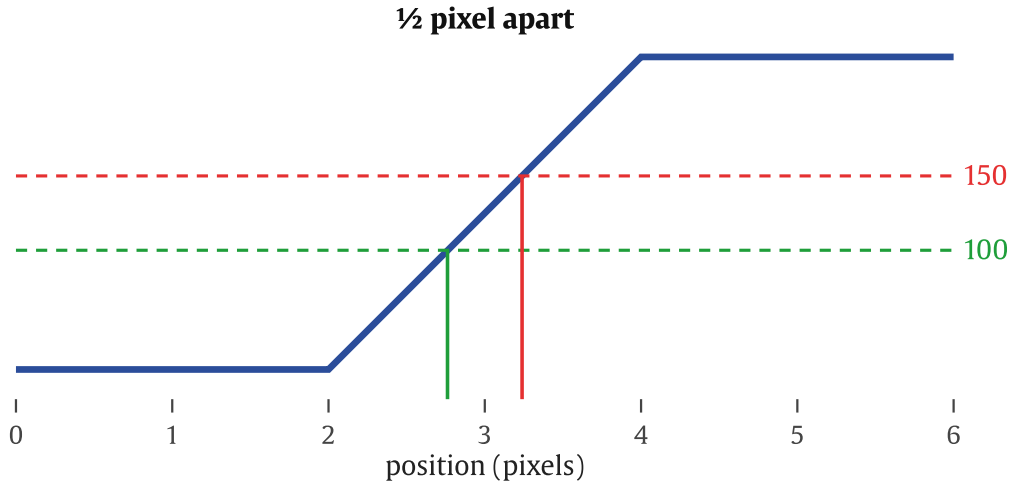
Parts and defects

Thresholding · Cleaning up · Blobs and their features

Thresholding

From grey levels to part and background — and why the threshold changes the size

- Every pixel above the threshold becomes 1, every pixel below it 0: a **binary** image.
- **Global** threshold: one value, from the valley of the histogram. **Otsu's method** finds it automatically: the value that separates the two groups of grey levels best.
- **Local (adaptive)** threshold: a different value in each region, when the light is not even.



THE THRESHOLD IS PART OF THE MEASUREMENT

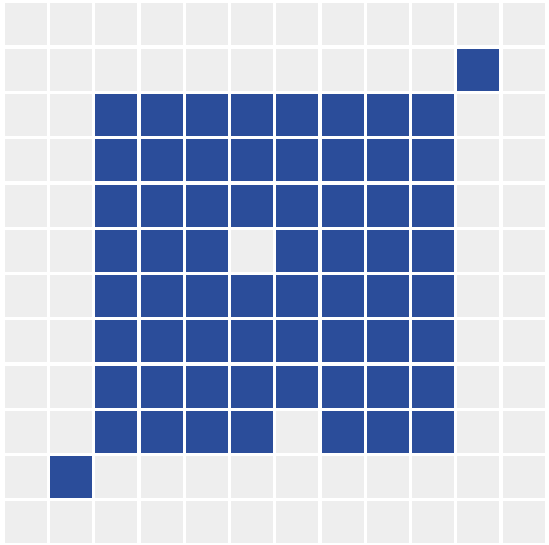
The washer's edge spreads over about two pixels, from 20 to 230. Moving the threshold from 100 to 150 moves each edge by half a pixel: the diameter changes by **one pixel, 41 µm** — a fifth of the hole's tolerance.

So

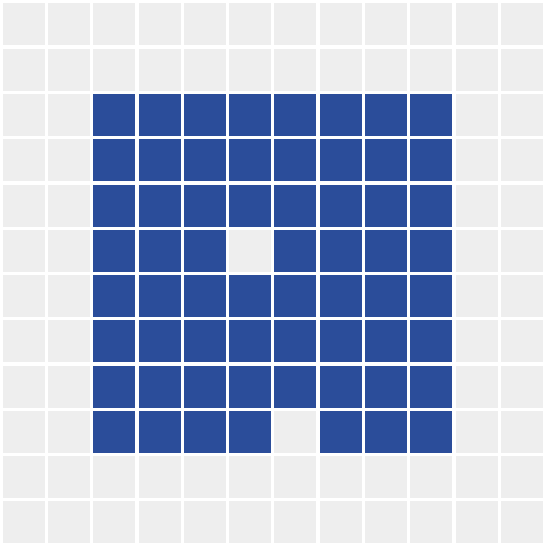
Fix the threshold, never let it float from image to image, and use the same one when calibrating on the master washer. The calibration then absorbs its effect.

Cleaning binary images

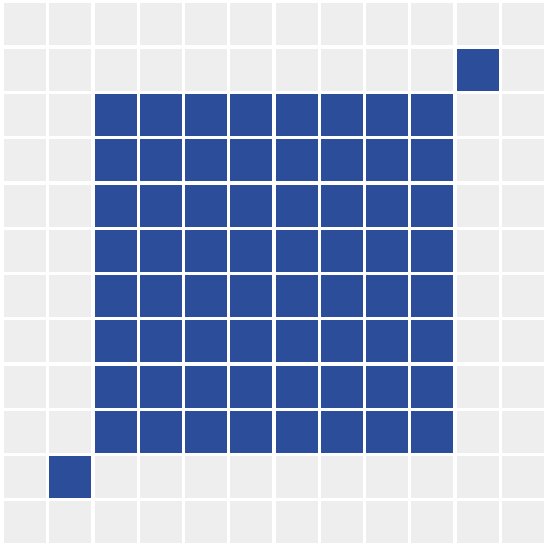
Removing specks and filling gaps, without changing the size



original: specks, a hole, a notch



opening: specks removed



closing: hole and notch filled

Erosion
Removes a layer of pixels from every boundary. Small specks vanish — and the part shrinks.

Dilation
Adds a layer. Small holes and gaps close — and the part grows.

Opening
Erosion, then dilation: **specks removed**, the part keeps its size.

Closing
Dilation, then erosion: **holes and cracks filled**, the part keeps its size.

Careful in defect inspection: an opening that removes dust also removes a defect of the same size. Size the clean-up below the smallest defect you must find.

Blobs and their features

Each connected group of pixels is one object, with numbers that describe it

Feature	How	Tells you
Area	number of pixels × area of one pixel	size; equivalent diameter $2\sqrt{(A/\pi)}$
Centroid	mean of the pixels' coordinates	position, to sub-pixel precision
Bounding box	smallest and largest row and column	quick size, touching the image edge?
Orientation	axis around which the pixels spread least	rotation of the part
Circularity	$4\pi A / P^2$, with P the perimeter	1 for a circle, small for a scratch
Elongation	spread along the main axis / across it	round or long
Holes	background regions inside the blob	is the washer's hole there?

THE WASHERS

Each washer is one blob with one hole. Its area uses **every** pixel of the part, so the equivalent diameter averages the noise of hundreds of edge pixels — finer than one pixel (lecture 8). Blobs touching the image border are discarded: they are cut.

THE STEEL COVER

Under dark field, scratches and dust both shine. A scratch is **long** (high elongation, low circularity); dust is **small and round**. Features tell them apart.

From features to limits

An inspection is a set of tolerances on image features

Defect on the cover	Rule
scratch	elongation > 5 and length > 0.5 mm
dent	area > 0.2 mm² and circularity > 0.6
dust (ignore)	area < 0.05 mm²

Illustrative limits — real ones come from the customer's surface specification and from real defects.

A defect catalogue

Limits are set on **real defects**: a collection of parts with every defect type, at every severity, labelled by experts — a defect atlas. Without it, nobody knows what the system should reject.

When rules are not enough

Defects that are hard to describe with a few features — textures, cosmetic flaws — are classified by models trained on many labelled images. They still need the same atlas, and the same validation.

PART 4

Finding and deciding

Template matching · Codes · Pass or fail · Validation

Template matching

Where in the image does this pattern appear?

- A small reference image — the **template** — slides over the image; at every position, a similarity score.
- **Normalised cross-correlation** (NCC): 1 for a perfect match, 0 for no relation. It ignores changes of brightness and contrast, so the score survives a dimmer lamp.
- The best score gives the **position**, to sub-pixel precision by interpolating around the peak.

Limits

A template matches only at the size and rotation it was taken. Parts that turn or change scale need many templates — or features that do not care: **interest points**, lecture 13.

Presence and absence

Is the label, the clip, the O-ring there? A score below a limit means missing.

Alignment before measuring

Find the part first, then place the measurement windows relative to it. Most gauging programs start this way.

Matching two views

The same similarity finds corresponding points in two cameras: the basis of stereo 3D measurement (lecture 14).

Reading codes and characters

Traceability: which part, from which batch, made when

- **1D barcodes** and **2D codes** (Data Matrix, QR) carry a serial or batch number. 2D codes include error correction: they read even when partly damaged.
- **Direct part marking:** codes engraved, laser-marked or dot-peened on the part itself, so the identity survives the whole life of the part — read with the right light (coaxial or dark field, lecture 10).
- **Character recognition** (OCR) reads printed dates and lot numbers; **verification** (OCV) checks that a known text is printed correctly.

Print quality is graded too

A code that reads today in the factory must still read at the customer. ISO/IEC 15416 (1D) and ISO/IEC 15415 (2D) grade the print quality from A to F — contrast, damage, distortion — with a verifier, which is itself a calibrated camera.

Why it matters for quality

A failure found in the field can be traced to its batch, its line and its shift — and the containment limited to the parts that are really affected.

Pass or fail: two ways to be wrong

Every inspection has a false-reject rate and a false-accept rate

	Part is good	Part is bad
System accepts	correct	**false accept — an escape**
System rejects	**false reject** — scrap	correct

- Moving a limit trades one error for the other: tighter limits, fewer escapes, more good parts scrapped.
- For measured values, the uncertainty decides near the limits: a **guard band** (ISO 14253-1) states which side the risk falls on — you apply it in the lab's third exercise.

What each error costs

A false reject is an **internal failure** cost: scrap and rework. A false accept is an **external failure** cost: complaints, returns, recalls, and the customer's trust — usually far more expensive (the cost of quality, lecture 4).

Rejected is not the end

Keep the image of every rejected part. Re-inspected by people, rejects tell you the false-reject rate for free — and show when the process, not the inspection, is drifting.

Validating a vision inspection

A camera system is a measuring instrument: it is proven before it is trusted

Inspection for defects (attributes)

A test set of parts whose status is known — good, bad, and above all **borderline** — each inspected many times. Report the detection rate and the false-reject rate, and how often the system agrees with itself and with the experts.

Measurement (variables)

A **gauge study**, as for any gauge: repeatability (same part, many images), reproducibility (different operators, setups, days), and bias against a reference instrument. The lab does a small one.

ISO 9001 asks for it

§7.1.5: monitoring and measuring resources must be suitable, kept suitable, and — where traceability is required — calibrated against standards. A vision system is one of them.

Re-validate after every change

A new lamp, lens, camera position or software version is a new instrument. Keep a set of reference parts and images, and run them after every change — and at every shift start, as a quick check.

Recap TAKEAWAYS

Parts, defects, finding and deciding

1. **Threshold:** from the histogram's valley or Otsu. It moves edges on blurred images — fix it, and calibrate with it.
2. **Clean-up:** opening removes specks, closing fills gaps, both keep the size — but never remove what is the size of a defect.
3. **Blobs:** area, centroid, orientation, circularity, holes. Limits on features are the inspection's tolerances, set on a defect catalogue.
4. **Matching and codes:** NCC finds parts and patterns; codes give traceability, with graded print quality.
5. **Deciding and validating:** false accepts against false rejects, guard bands near limits, a gauge study before use, re-validation after every change.

The washers, processed

The whole pipeline on one case — the lab's second and third exercises

Step	What is done	What could go wrong
Acquire	backlight, fixed exposure, lossless image	saturated background, JPEG
Check	histogram: two peaks, nothing at 0 or 255	low contrast, clipping
Segment	fixed threshold in the valley; opening to remove dust	a floating threshold changes the size
Find the parts	blobs with one hole; drop those cut by the border	touching washers seen as one
Measure	equivalent diameter from the area; circles fitted to the edges	perspective, distortion
Calibrate	mm per pixel from the master washer, same threshold	master measured badly
Decide	ISO 7089 limits with a guard band	ignoring the uncertainty
Record	values, images of rejects, settings	no traceability

Next
2D dimensional measurement: calibration, sub-pixel gauging and its accuracy, and in-line systems.

Then
Interest points: finding the same feature in two images, whatever its size, rotation and light.

Questions?

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github.com/LolloCappo/student-hub